

Stable-Plastic Context Memory for Dataset-Agnostic Continual Routing in MMALS

A Backward-Compatible Architectural Proposal and Reviewer-Facing Evaluation Protocol

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Abstract

The latest CORE50 experiments expose a structural mismatch in the current MMALS context path. A v2.3 rescue ablation improves average accuracy, forgetting, class coverage and host diversity, yet fails to recover the two oldest contexts. The isolated exponential-moving-average intervention also fails. Inspection shows that the retention objective directly trains a single-input context encoder, whereas the deployed posterior is dominated by prototype evidence. This note proposes a dataset-agnostic stable-plastic context-memory module that trains the posterior actually used at inference, preserves slow context anchors while allowing plastic updates, and provides an exact **legacy** operating mode for reproducing all prior notebooks. The module is designed to work after any input representation, including raw-flat MNIST-family inputs and frozen visual features for SplitCIFAR or CORE50. The proposal includes three backward-compatible modes, stable and plastic memory banks, adaptive fusion, teacher-posterior and functional distillation, overlap-aware soft targets, a confident-collapse veto, computational controls, an ablation matrix, falsification criteria and cross-dataset regression gates. This is an architectural proposal and test plan, not evidence that the redesign already improves performance.

1 Reviewer-facing motivation

The proposal is driven by a falsified hypothesis rather than by architecture inflation. In the two-seed CORE50 v2.3 ablation, the combined rescue raises final average accuracy from 0.2734 to 0.3171, minimum-task accuracy from 0.0050 to 0.0513, reduces mean zero-recall classes from 9.0 to 0.5, and increases effective hosts from about 2.04 to 8.68. Nevertheless, oldest-two context top-1 remains 0.0. The isolated context EMA intervention reduces oldest-two task accuracy and increases zero-recall classes. Therefore, simply preserving prototypes more strongly is not sufficient.

The relevant mismatch is:

$$\text{training objective} \longrightarrow q_{\text{single}}, \quad \text{deployed decision} \longrightarrow q_{\text{prototype}} \text{ dominated.} \quad (1)$$

When the loss optimizes a weakly weighted branch and the dominant inference branch is only indirectly affected, the system can report improved intermediate-context behavior while the oldest contexts remain unrecovered.

This note asks a narrow question:

Can a common context-memory layer preserve old contextual evidence, remain plastic under drift, support legitimate ambiguity, and reproduce historical results exactly when disabled?

2 Design requirements

A reviewer should be able to reject the module if any of the following requirements are not met.

1. **Dataset agnosticism.** No branch may depend on dataset names, fixed class IDs, “old tasks 0 and 1,” or a fixed number of contexts or hosts.
2. **Backward compatibility.** A **legacy** mode must reproduce the historical deployed posterior and permit old checkpoints to be evaluated without recovering every historical notebook.
3. **Deployed-posterior training.** Retention losses must act on the posterior used during inference, not only on a subordinate component.

4. **Stable-plastic separation.** Slow anchors and rapidly updated prototypes must be explicit, separately audited objects.
5. **Ambiguity preservation.** Overlapping contexts must support soft targets and guarded mixed-context routing.
6. **Bounded complexity.** The module should add modest state and computation relative to the existing router, with exact cost reporting.
7. **No test leakage.** Fusion calibration, repair and selection must use train, repair or policy-validation memory only.

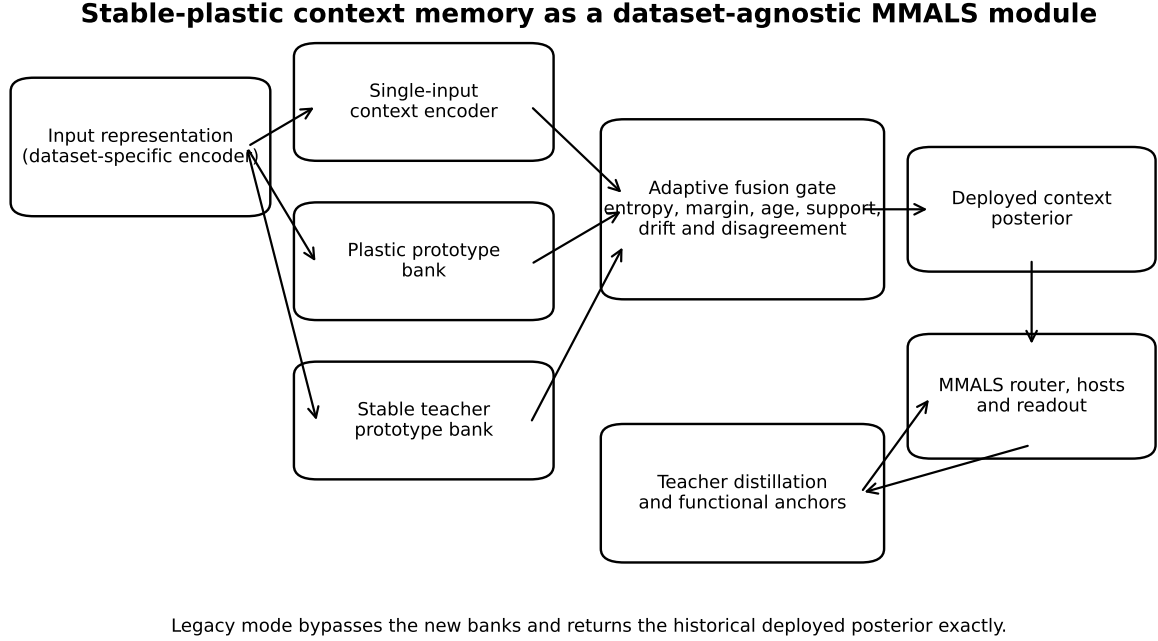


Figure 1: Proposed dataset-agnostic context-memory module. The input encoder remains dataset-specific; the context-memory interface is common.

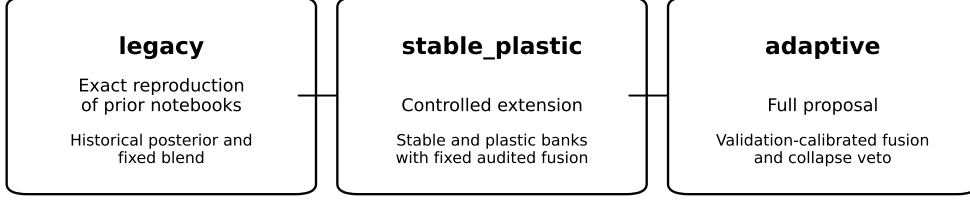
3 Backward-compatible operating modes

The module must expose an explicit mode contract rather than silently changing the behavior of MMALS.

Table 1: Operating modes and guarantees.

Mode	Purpose	Context posterior	Required guarantee
legacy	Exact historical reproduction	Existing single/prototype posterior and fixed blend	Bitwise or tolerance-bounded equivalence on stored checkpoints and test fixtures.
stable_plastic	Controlled architectural ablation	Single encoder plus stable and plastic prototype banks under fixed fusion	No dataset-specific branches; all weights and updates exported.
adaptive	Proposed deployable mode	Validation-calibrated fusion with disagreement and confident-collapse veto	Regression gates on MNIST family and SplitCIFAR10 before promotion.

Backward-compatible operating modes



Every dataset and prior result remains reproducible through the legacy path.

Figure 2: The proposal is an extension of the existing context path, not an irreversible fork.

In **legacy** mode, the module must implement

$$q_{\text{deploy}}^{\text{new}}(c \mid x, m) = q_{\text{deploy}}^{\text{historical}}(c \mid x, m) \quad (2)$$

within a declared numerical tolerance. The release package includes a reference module and tests for this contract.

4 Stable and plastic memory banks

For each context c , the module maintains two memory states.

The *plastic* prototype $P_c^{(t)}$ follows current repair memory and recent observations:

$$P_c^{(t)} \leftarrow (1 - \eta_p)P_c^{(t-1)} + \eta_p \hat{P}_c^{(t)}, \quad (3)$$

where η_p may be relatively large.

The *stable* anchor $S_c^{(t)}$ changes slowly and only when support and validation evidence are sufficient:

$$S_c^{(t)} \leftarrow (1 - \eta_s)S_c^{(t-1)} + \eta_s \hat{S}_c^{(t)}, \quad \eta_s \ll \eta_p. \quad (4)$$

Each bank records support count, age, last update, dispersion and drift. Stable memory is not “frozen forever.” It is slow, evidence-gated memory. Plastic memory is not trusted by default; it provides adaptation evidence.

For a sample representation e and mediated latent z , the module computes branch posteriors

$$q_s(c \mid e, z), \quad q_p(c \mid e, z), \quad q_0(c \mid e), \quad (5)$$

for stable, plastic and single-input evidence respectively.

5 Adaptive fusion

The proposed deployed posterior is

$$q_{\text{deploy}}(c \mid x, m) = \alpha_0 q_0(c \mid e) + \alpha_p q_p(c \mid e, z) + \alpha_s q_s(c \mid e, z), \quad (6)$$

where $\alpha_0 + \alpha_p + \alpha_s = 1$ and $\alpha_i \geq 0$.

The fusion weights are functions of auditable features:

$$\alpha = g(H(q_0), H(q_p), H(q_s), \Delta_{sp}, \gamma, a, n, d), \quad (7)$$

where H is entropy, Δ_{sp} is stable-plastic disagreement, γ is the top-two margin, a is context age, n is support and d is drift. The first implementation should support both a rule-based gate and a small learned gate. The rule-based gate is preferred for the first causal ablation because its decisions are easier to audit.

The adaptive mode must permit three limiting cases:

$$\alpha = (1, 0, 0) \quad \text{single-input fallback,} \quad (8)$$

$$\alpha = (0, 1, 0) \quad \text{plastic adaptation,} \quad (9)$$

$$\alpha = (0, 0, 1) \quad \text{stable-anchor protection.} \quad (10)$$

It must also permit the exact historical blend in `legacy` mode.

6 Train the posterior used at inference

For replayed old-context samples (x, c) , the first retention term is direct cross-entropy on the deployed posterior:

$$\mathcal{L}_{\text{ctx}} = -\log q_{\text{deploy}}(c \mid x, m). \quad (11)$$

A teacher snapshot from the previous checkpoint provides a soft retention target:

$$\mathcal{L}_{\text{ctxKD}} = T^2 D_{\text{KL}}(q_{\text{deploy}}^*(\cdot \mid x, m; T) \parallel q_{\text{deploy}}(\cdot \mid x, m; T)). \quad (12)$$

This term preserves uncertainty where uncertainty was previously legitimate. It does not force a hard context label when the teacher distribution was mixed.

7 Functional preservation beyond context labels

Correct context identity is useful but insufficient. The CORE50 oracle gap shows that even perfect context would leave substantial classification error. The old-state objective should therefore preserve a compact functional bundle:

$$\mathcal{L}_{\text{old}} = \lambda_c \mathcal{L}_{\text{ctxKD}} + \lambda_r \|r - r^*\|_2^2 + \lambda_z \|z_f - z_f^*\|_2^2 + \lambda_y T_y^2 D_{\text{KL}}(p_y^* \parallel p_y). \quad (13)$$

This formulation follows the earlier MMALS finding that route preservation alone is not equivalent to preservation of the mediated latent function.

To limit memory cost, teacher bundles can be stored for a small balanced replay subset or reconstructed from periodic checkpoint anchors. The implementation must report bytes per context and bytes per replay item.

8 Overlap-aware targets

The MNIST-family overlap chain contains legitimate ambiguity. For example, a shared class may support two adjacent contexts. A hard one-hot context target can therefore destroy a useful mixed posterior.

For overlapping contexts, the target can be a normalized support distribution

$$q^*(c \mid x) \propto \mathbf{1}[c \in \mathcal{C}(y)] w_c(x), \quad (14)$$

where $\mathcal{C}(y)$ is the set of contexts containing class y and w_c may use teacher evidence or prototype distance. This is a protocol option derived from task metadata, not a dataset-name condition.

Table 2: Compatibility implications across current benchmark families.

Dataset family	Context structure	Design consequence
MNIST / FashionMNIST overlap chain	Adjacent tasks share classes and ambiguity is legitimate.	Use soft teacher targets and permit guarded mixed-context posteriors; do not force hard prototype separation.
Rotated / Permuted MNIST	Contexts reflect transformation or domain changes.	Stable anchors preserve transformation identity while the plastic bank adapts to drift.
SplitCIFAR10	Five disjoint class groups with a strong current result.	New module must be opt-in and pass strict non-regression gates before becoming default.
SplitCIFAR100	Ten large class groups with representation/readout limits.	Context redesign may help retention but cannot substitute for stronger representation and readout controls.
CORE50	Ten object groups with observed recency and host collapse.	Primary failure-driven test bed for stable-plastic memory and deployed-posterior distillation.

9 Confident-collapse veto

The CORE50 failure was not merely high entropy. Old samples were sometimes mapped to a recent context with near-zero entropy. Therefore, low entropy cannot be interpreted as correctness.

Define the predicted context age gap $\Delta a = a(\hat{c}) - a(c_s)$, where c_s is the stable-anchor winner. A confident recent-context collapse alert is triggered when

$$H(q_{\text{deploy}}) < h_{\min}, \quad \Delta a > a_{\max}, \quad q_s(c_s) - q_s(\hat{c}) > \delta_s. \quad (15)$$

The response is not always to force c_s . Depending on validation configuration, the module may:

1. increase stable-anchor weight;
2. use a guarded mixture of the stable and single branches;
3. route through multiple contexts;
4. abstain to a safe policy family.

Every intervention must be exported with reason codes.

10 Dataset-agnostic interface

The module operates after the input representation. Its required dimensions derive from configuration:

$$N_{\text{contexts}}, \quad d_e, \quad d_z, \quad N_{\text{hosts}}, \quad \text{support metadata}. \quad (16)$$

It must not inspect dataset names or class literals. The same interface applies to raw-flat MNIST inputs, frozen CNN features, future graph encoders or other representations.

A minimal forward contract is:

```
posterior, audit = context_memory(  
    q_single=q_single,  
    stable_logits=stable_logits,  
    plastic_logits=plastic_logits,  
    metadata={entropy, margin, age, support, drift, disagreement},  
    mode="legacy|stable_plastic|adaptive",  
)
```

The returned audit object includes fusion weights, branch winners, disagreement, collapse-veto status and fallback reason.

11 Computational and memory cost

For C contexts and prototype dimension d , two prototype banks add approximately $2Cd$ floating-point values plus metadata. With $C = 10$ and $d = 256$, this is negligible relative to the host backbone. The learned adaptive gate should remain small, for example a two-layer MLP over branch diagnostics, not over raw images.

The expensive part is functional teacher memory. The first implementation should compare:

1. posterior-only distillation;
2. posterior plus route anchors;
3. posterior, route and latent anchors;
4. full output distillation.

The smallest variant passing retention gates should be preferred.

12 Ablation matrix

The redesign must not be evaluated only as one combined “full rescue.” Each hypothesis is isolated.

Table 3: Required mechanism ablations.

Variant	Isolated question	Promotion role
legacy	Can previous results be reproduced without old notebooks?	Mandatory reference
current_ema	Does the v2.3 EMA intervention reproduce its observed failure?	Failure control
adaptive_blend_only	Was fixed prototype dominance the main mismatch?	Mechanism isolation
stable_plastic_only	Do slow anchors prevent prototype drift?	Mechanism isolation
deployed_posterior_distill	Does training the actual inference posterior retain old contexts?	Mechanism isolation
stable_plastic_plus_distill	Do the two corrections interact positively?	Candidate module
oracle_blend_diagnostic	What is the achievable context ceiling without deployment leakage?	Diagnostic only

Each branch must export separate top-1 accuracy, entropy and confusion for

$$q_0, \quad q_{\text{feature}}, \quad q_{\text{latent}}, \quad q_p, \quad q_s, \quad q_{\text{deploy}}. \quad (17)$$

Results must also be grouped by context age: oldest two, early-middle, late-middle and newest two.

13 Promotion and regression gates

Table 4: Initial promotion gates. Thresholds are preregistered engineering gates, not universal constants.

Gate	Required value
Oldest-two context top-1 gain	$\geq +0.20$
Oldest-two task accuracy gain	$\geq +0.03$
Zero-recall class reduction	≥ 2
Final average accuracy regression	≤ 0.02
Recent-two task regression	≤ 0.03
Confident recent-context collapse	Strongly reduced
SplitCIFAR10 final accuracy regression	≤ 0.01

The first mechanism test remains COrE50 seeds 0 and 4. If the candidate passes, it proceeds through the regression chain shown below.

Promotion sequence: mechanism before broad claim



A failure at any stage returns the module to ablation rather than silently changing the default.

Figure 3: Cross-dataset promotion sequence. The qualified SplitCIFAR10 result is protected before the redesign becomes a default backbone component.

MNIST-family regression must preserve qualitative selector behavior: clean MNIST should not be over-guarded, FashionMNIST ambiguity should remain protected, PermutedMNIST should retain simple/global options, and RotatedMNIST should retain weak-boundary calibration when validation evidence supports it.

14 Reviewer questions and planned answers

Is the proposal merely more capacity?

No claim will be accepted without parameter, memory and latency deltas. The key mechanism variants share the same host backbone. Legacy, fixed stable-plastic and adaptive fusion are compared at matched representation and host count before integration with larger features or additional hosts.

Does the stable bank just freeze old tasks?

No. Stable anchors are slowly updated and evidence-gated. The module permits shared contexts and soft targets. The ablation compares frozen, slow and plastic updates.

Could the module overfit COrE50?

The interface contains no COrE50 literals. Promotion requires MNIST-family and SplitCIFAR10 regression. SplitCIFAR100 is used as a harder diagnostic before returning to five-seed COrE50 evidence.

Does teacher distillation prevent learning new contexts?

Plasticity is measured explicitly through newest-task and recent-two-task accuracy. A candidate fails if recent performance regresses beyond the preregistered bound.

Can previous results still be reproduced?

Yes, only if legacy-equivalence tests pass. Historical checkpoints should load through a versioned adapter that reconstructs missing stable/plastic state as disabled tensors and returns the original posterior.

Does the proposal prove backward transfer?

No. The intended claims are retention, context stability, reduced collapse and bounded plasticity loss. Positive backward transfer remains a separate hypothesis.

15 Falsification criteria

The redesign should be rejected or delayed if:

1. it fails to raise oldest-context accuracy on seeds 0 and 4;
2. gains disappear when representation and host count are matched;
3. legacy mode cannot reproduce stored outputs;
4. MNIST-family ambiguity is harmed by hard context separation;
5. SplitCIFAR10 regresses beyond the declared limits;
6. improvements require disproportionate memory or inference latency;
7. the adaptive gate becomes a non-auditable dataset classifier;
8. the module improves context top-1 but not old-task or class retention.

16 Implementation and release contract

A versioned checkpoint should include:

- context-memory schema version;
- operating mode;
- legacy blend parameters;
- stable and plastic prototype banks;
- support, age, dispersion and drift metadata;
- fusion-gate state;
- teacher-anchor configuration;
- hash of the representation and router configuration.

The companion GitHub package includes a reference PyTorch module and unit tests for legacy equivalence, probability normalization, arbitrary context counts and absence of dataset-specific logic. The module is a design reference and is not yet integrated into the full training notebook.

17 Conclusion

The proposed stable-plastic module is a localized evolution of the MMALS backbone. It does not replace the hosts, transformations, mediated latent state, readout, audit layers or RC2O policy framework. It replaces a fixed and partly mismatched context-memory interface with a versioned, backward-compatible component that separates slow and fast memory, trains the deployed posterior, preserves legitimate ambiguity and exposes its decisions. The proposal is justified by a concrete failed promotion gate and is accompanied by a sequence that can disprove it before any broad claim is made.

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